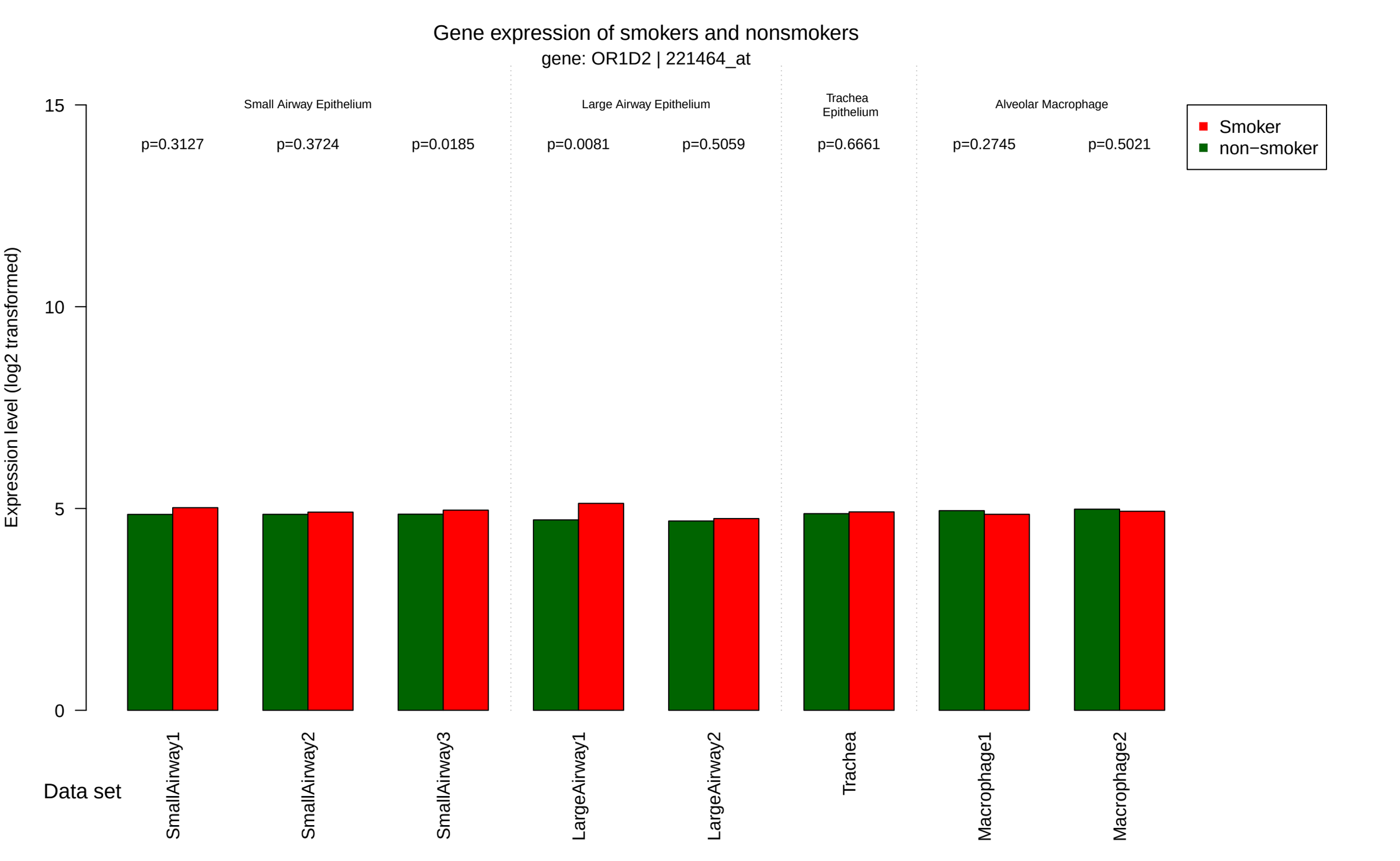
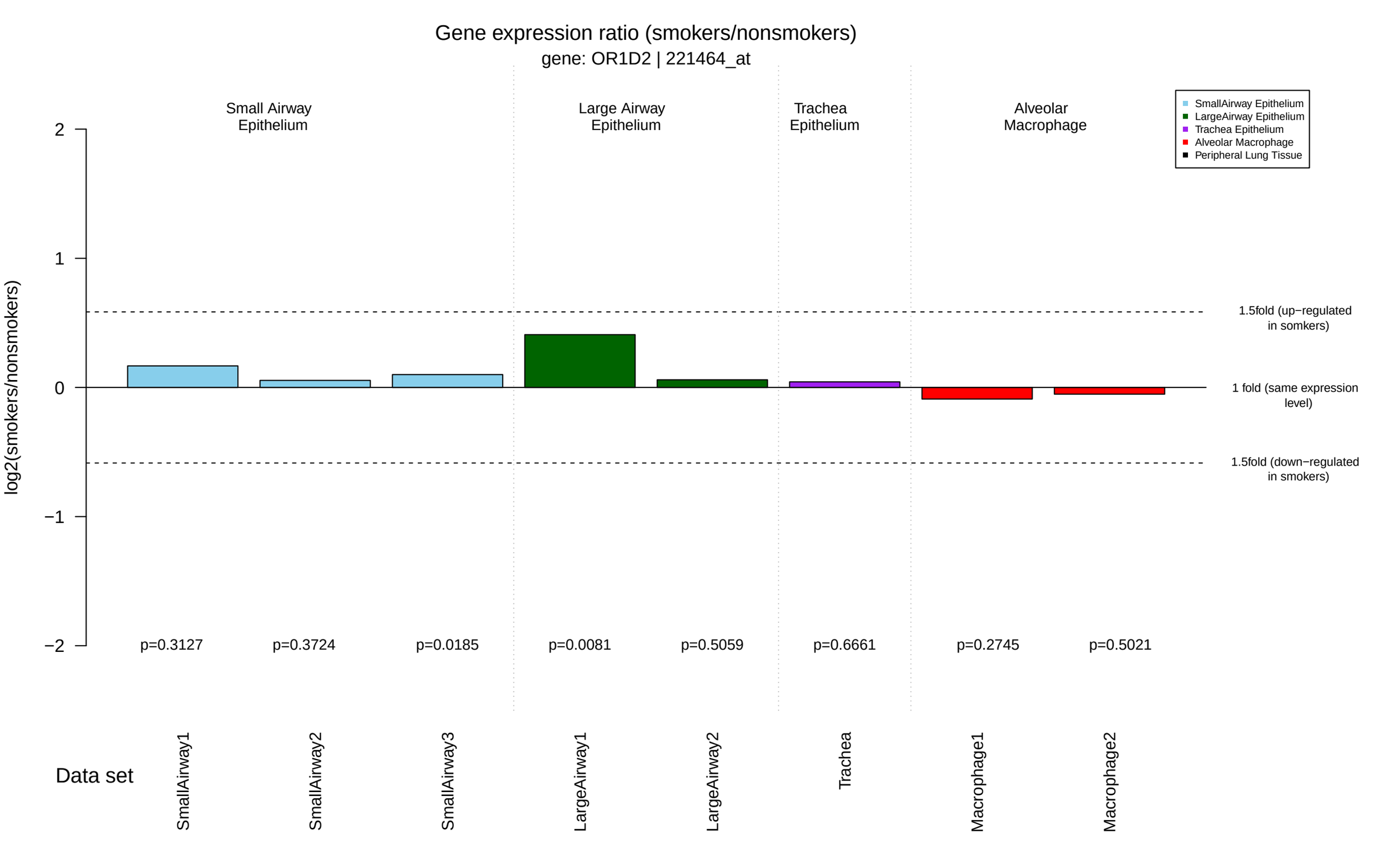
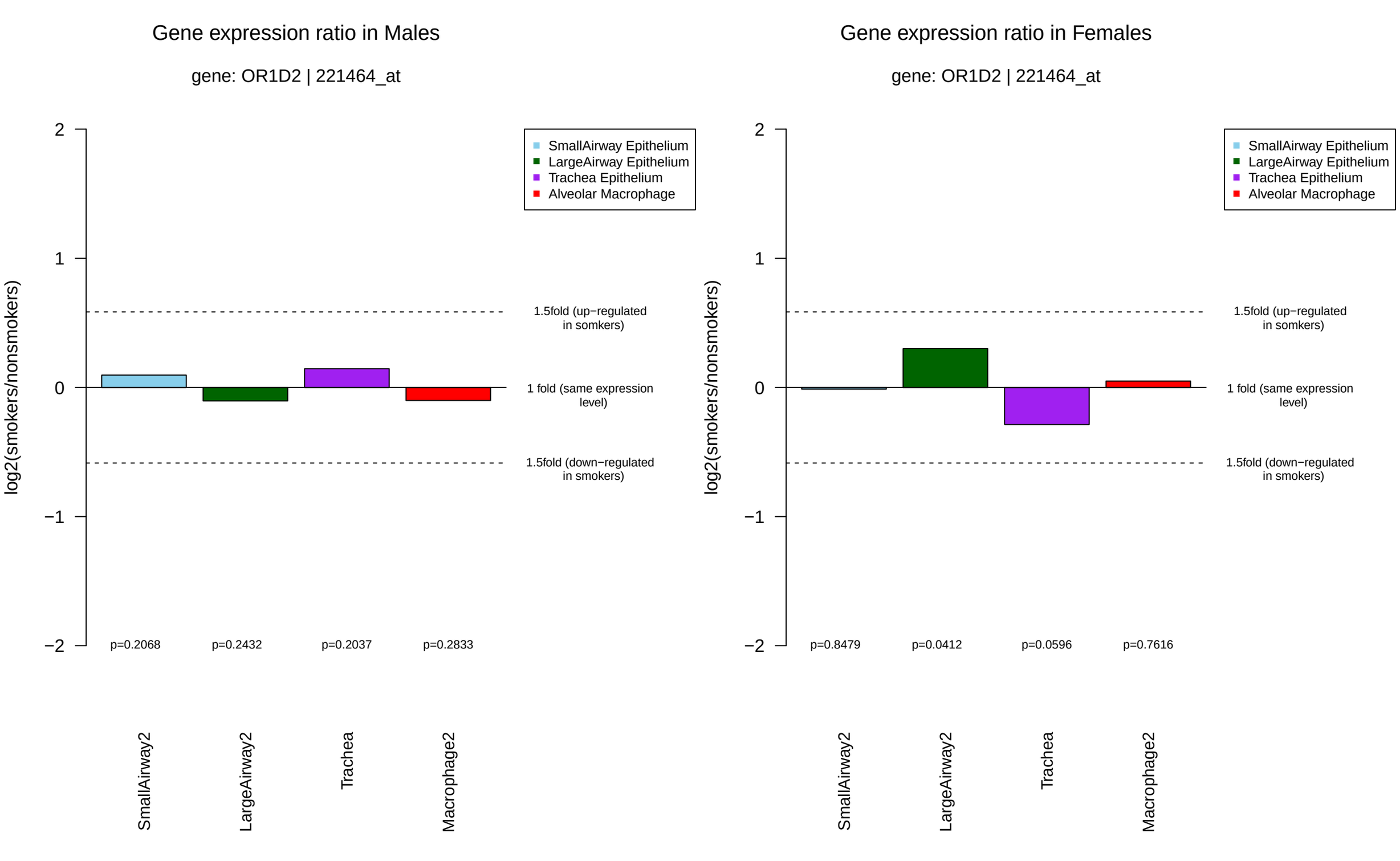
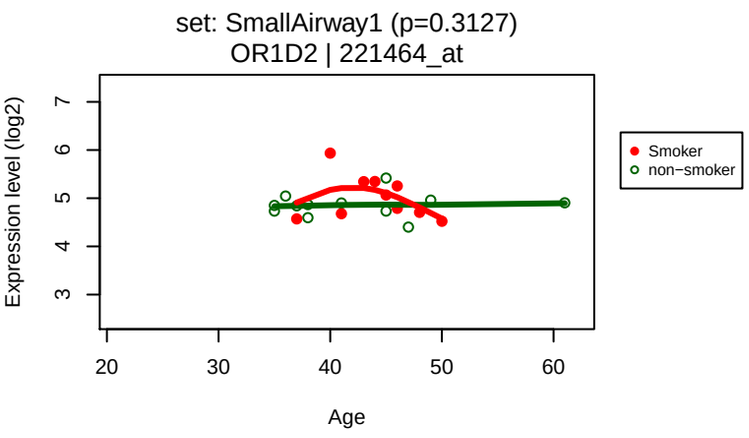
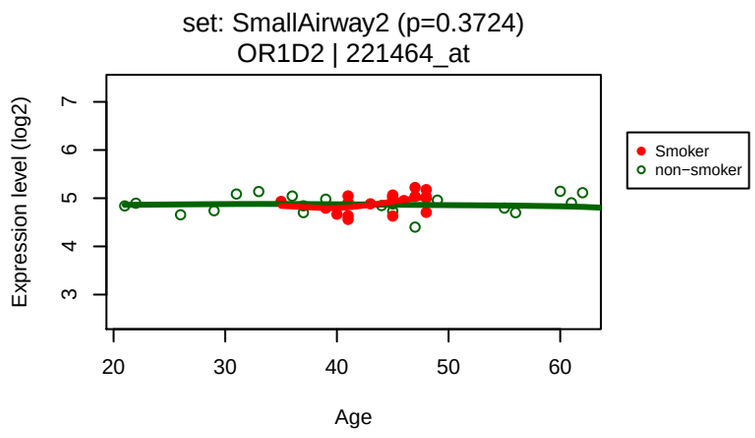
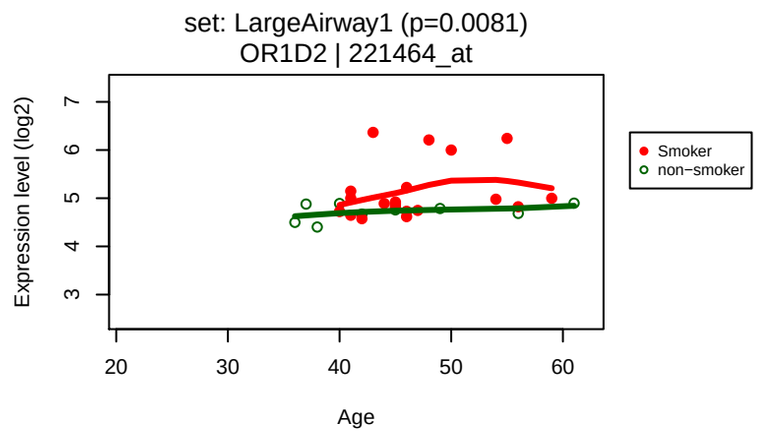

Gene expression of smokers and nonsmokers
gene: OR1D2 | 221464_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3127
p=0.3724
p=0.0185
p=0.0081
p=0.5059
p=0.6661
p=0.2745
p=0.5021
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

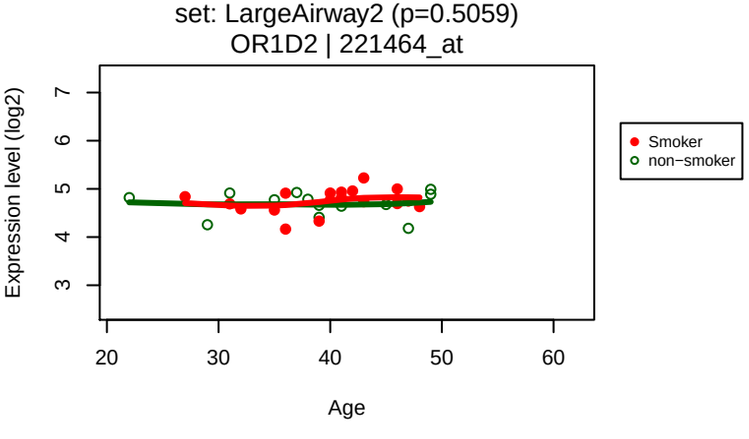

Gene expression ratio (smokers/nonsmokers)
gene: OR1D2 | 221464_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3127
p=0.3724
p=0.0185
p=0.0081
p=0.5059
p=0.6661
p=0.2745
p=0.5021
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

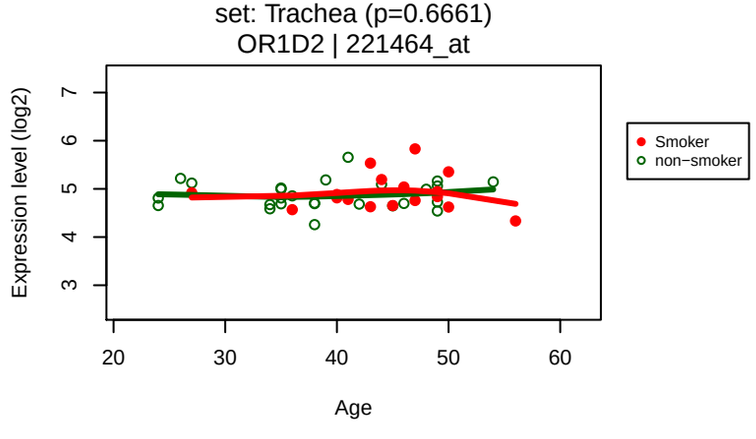

Gene expression ratio in Males
gene: OR1D2 | 221464_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2068
p=0.2432
p=0.2037
p=0.2833
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR1D2 | 221464_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8479
p=0.0412
p=0.0596
p=0.7616
SmallAirway2
LargeAirway2
Trachea
Macrophage2

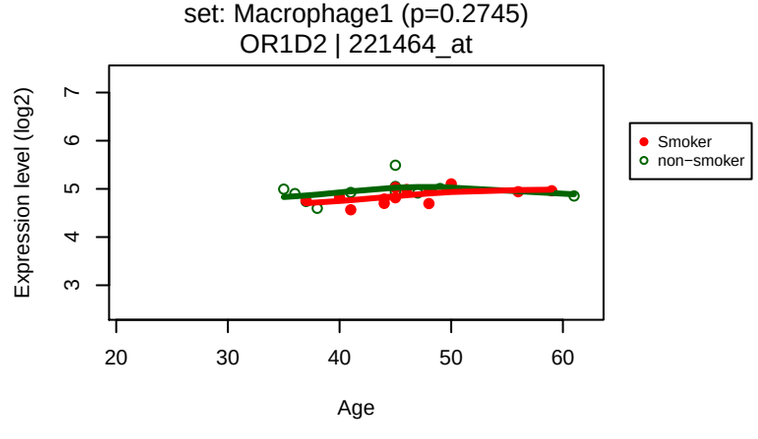

set: SmallAirway1 (p=0.3127)
OR1D2 | 221464_at
Expression level (log2)
Age
Smoker
non−smoker

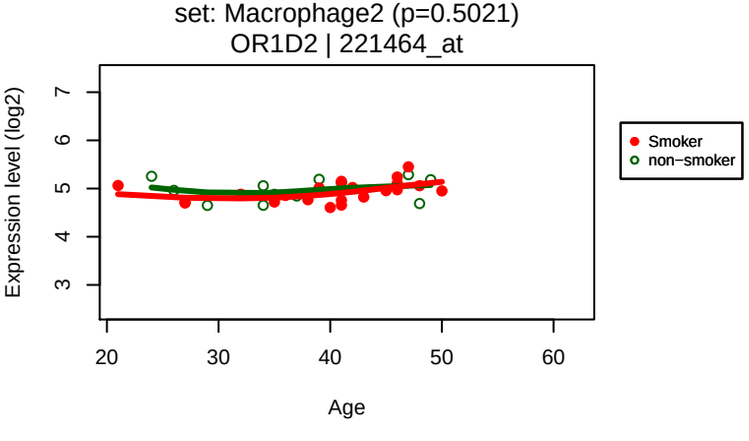

set: SmallAirway2 (p=0.3724)
OR1D2 | 221464_at
Expression level (log2)
Age
Smoker
non−smoker

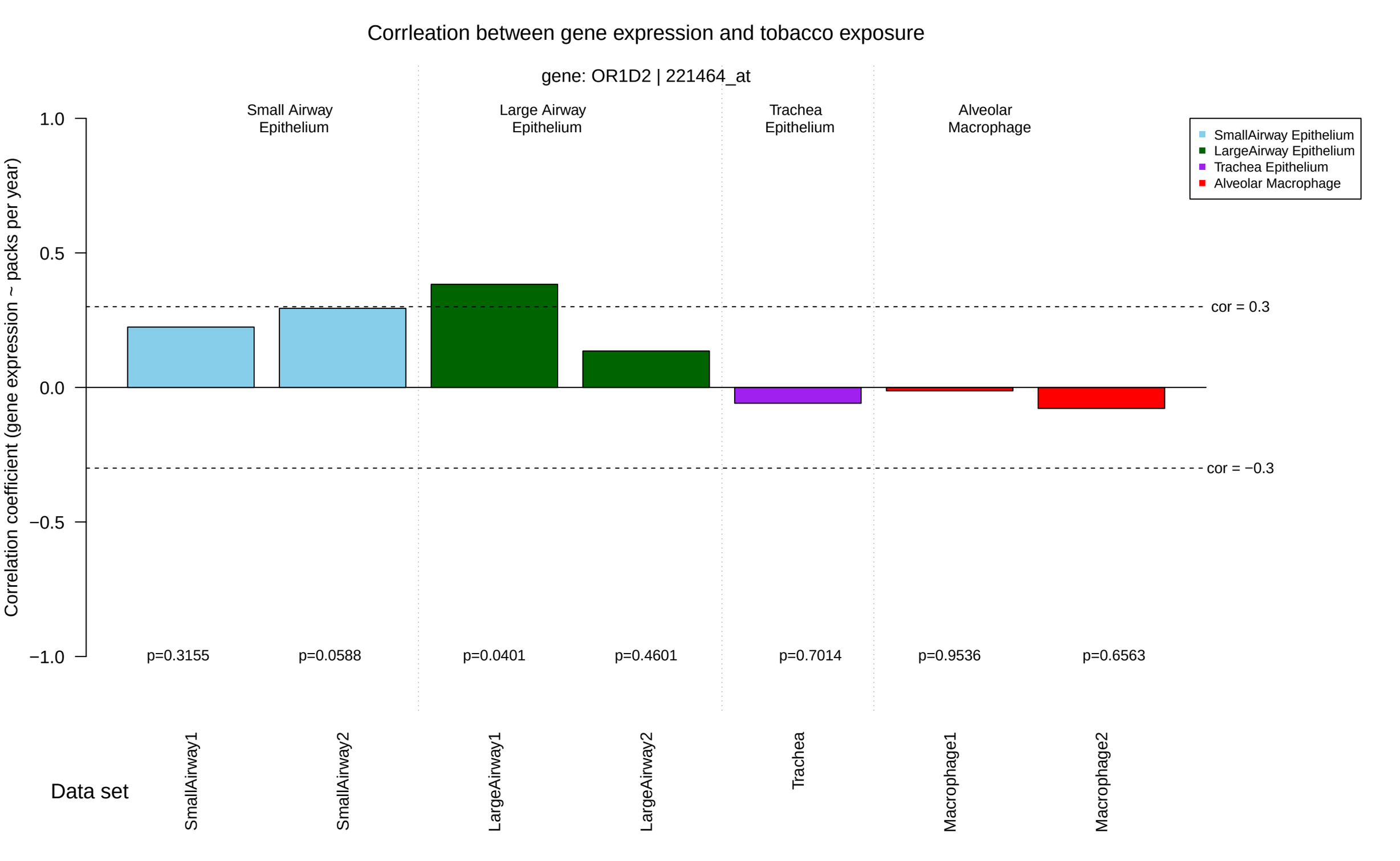

set: LargeAirway1 (p=0.0081)
OR1D2 | 221464_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5059)
OR1D2 | 221464_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6661)
OR1D2 | 221464_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2745)
OR1D2 | 221464_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5021)
OR1D2 | 221464_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR1D2 | 221464_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3155
p=0.0588
p=0.0401
p=0.4601
p=0.7014
p=0.9536
p=0.6563
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2